BUSINESS PROBLEM • TIME SERIES FORECASTING

CASE STUDY – FORECASTING TRACTOR SALES

Demand Forecasting for Production and Inventory Planning

1. Business Context

PowerHorse is a tractor and farm-equipment manufacturer that has experienced consistent revenue growth since its establishment. However, fluctuations in tractor sales and demand have made it difficult to control production and inventory costs. Producing too many tractors can increase holding costs and tie up working capital, while producing too few can lead to stockouts, delayed deliveries, and missed sales.

The company wants to improve production planning through reliable demand forecasts and also understand whether its marketing and farmer-connect activities are associated with changes in sales.

2. Business Problem Statement

How can PowerHorse forecast future monthly tractor demand accurately enough to support production and inventory decisions, reduce avoidable costs, and evaluate the relationship between marketing spend and tractor sales?

3. Business Objectives

- Analyse 12 years of monthly tractor sales to identify trend, seasonality, cycles, and unusual observations.
- Decompose the sales series into trend, seasonal, and residual components.
- Forecast tractor sales for the next four quarters for near-term planning and, where supported by the data and validation, extend the monthly forecast to a three-year horizon.
- Test and compare time-series and regression approaches, including ARIMA, SARIMA, ARIMAX, SARIMAX, and regression-based models.
- Use the sales-and-marketing dataset to assess whether marketing spend adds explanatory or predictive value to sales forecasts.
- Evaluate forecast accuracy using RMSE, MAE, and MAPE where appropriate, and select a model based on out-of-sample performance.

4. Data Available

Dataset	Coverage	Purpose
Tractor-Sales.csv	Monthly number of tractors sold over 12 years	Explore sales patterns, decompose the series, and build baseline and seasonal forecasts.
Sales-and-Marketing.csv	Monthly tractor sales and marketing spend over the last four years	Test models with an exogenous marketing-spend variable and examine its relationship with sales.

5. Proposed Analytical Approach

- **Data preparation:** Parse dates, sort observations chronologically, check missing values and duplicates, verify monthly frequency, and inspect outliers.
- **Exploratory time-series analysis:** Plot monthly sales and review trend, seasonality, variability, and potential structural changes.
- **Seasonal decomposition:** Use an appropriate additive or multiplicative decomposition to separate trend, seasonal pattern, and residual variation.

- **Model development:** Build a simple baseline and compare regression, ARIMA, SARIMA, ARIMAX, and SARIMAX as suitable for the data.
- **Validation and selection:** Use chronological train/test splits or rolling-origin validation; avoid random shuffling of time-series observations.
- **Forecasting and interpretation:** Generate forecasts with prediction intervals where available, compare errors, and translate the results into planning insights.

6. Evaluation and Expected Business Value

Forecast evaluation: RMSE (penalises larger errors), MAE (average absolute error), and MAPE (average percentage error when actual values are non-zero and suitable for percentage comparisons). The chosen model should be selected using held-out future periods, not training fit alone.

Expected business value: better-informed production schedules, lower risk of overproduction and stockouts, more efficient inventory and working-capital planning, and evidence to help assess whether marketing investment contributes useful predictive information about sales.

Important scope note: The brief mentions both a three-year forecast and a forecast for the next four quarters. This document treats the next four quarters as the near-term planning deliverable and a three-year monthly forecast as the longer-horizon objective. Final horizons should be confirmed with the project sponsor. Model accuracy and marketing impact cannot be concluded until the datasets are analysed.